

Thermal Performance Evaluation of Space Radiator for Single-Phase Mechanically Pumped Fluid Loop

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<https://doi.org/10.2514/1.A35030>

This paper presents the experimental and analytical results of a heat rejection system consisting of a single-phase mechanically pumped loop and a space radiator. It has been developed for future crewed exploration missions to provide a large amount of heat dissipation capability. The radiator consists of a honeycomb radiator, eight aluminum pipes inserted through the honeycomb core, and two aluminum manifolds joined at both ends of the eight pipes. A graphite sheet is attached to the back of the aluminum skin, facing deep space to improve the thermal diffusivity in the in-plane direction. Silverized Teflon® tapes are put on the surface of the skin on the deep-space side, and the surface of the opposite skin is covered with multilayer insulation. A gap between the radiator surface and the flow channel protects against micrometeoroids and orbital debris. A thermal vacuum test was conducted to evaluate the thermal performance of the radiator. The coolant was HFE7200, and its flow rate was varied from about 50 to 200 kg/h. More than 50 temperatures across the radiator surface were measured for each case. Thermal Desktop® and SINDA/FLUINT software were used to develop the thermal mathematical model, which was correlated with the test results and evaluated through thermal analysis. The fin efficiency of the radiator increased with the mass flow rate of the coolant, reaching 0.9 or higher.

Nomenclature

A_i	=	area of i th node, m^2
A_{rad}	=	radiator surface area, m^2
C_p	=	specific heat capacity, $J/kg/K$
c_b	=	bumper coefficient
c_w	=	rear wall coefficient
d_p	=	projectile diameter, cm
d_t	=	tube diameter, cm
h	=	heat transfer rate between coolant and tube wall, $W/m^2/K$
k	=	coolant thermal conductivity, $W/m/K$
M_p	=	projectile mass, g
m_p	=	projectile areal density, g/cm^2
$\dot{m}$	=	mass flow rate, kg/h
Nu	=	Nusselt number
Nu_{lam}	=	Nusselt number of laminar flow
Nu_{tra}	=	Nusselt number of transitional flow
Nu_{tu}	=	Nusselt number of turbulent flow
Pr	=	Prandtl number
Q_{id}	=	identical heat dissipation amount, W
Q_{leak}	=	heat leak, W

Q_{leak_jig}	=	heat leak amount though mounting jig of radiator panel, W
Q_{leak_MLI}	=	heat leak amount though multilayer insulation attached on backside of radiator panel, W
Q_{leak_mnf}	=	heat leak amount though manifolds, W
Q_{leak_side}	=	heat leak amount though sides of radiator panel, W
Q_{real}	=	actual heat dissipation amount, W
Q_{real_a}	=	actual heat dissipation amount calculated from analysis results, W
Q_{real_t}	=	actual heat dissipation amount calculated from test results, W
Re	=	Reynolds number
r_i	=	inner radius of tube, m
r_o	=	outer radius of tube, m
r^*	=	ratio of outer radius and inner radius
S	=	overall spacing between outer bumper and rear wall, cm
T_i	=	temperature of i th node, K
T_{in}	=	inlet coolant temperature, K
T_{max}	=	maximum radiator surface temperature, K
T_{out}	=	outlet coolant temperature, K
T_{sh}	=	shroud temperature, K
t_b	=	bumper thickness, cm
t_w	=	rear wall thickness, cm
V	=	projectile velocity, km/s
V_n	=	normal component of projectile velocity, km/s
ϵ	=	hemispherical infrared emissivity
η	=	fin efficiency
ρ_b	=	bumper density, g/cm^3
ρ_p	=	projectile density, g/cm^3
σ_s	=	Stefan–Boltzmann constant, $W/m^2/K^4$
σ_y	=	rear wall yield stress, ksi

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I. Introduction

SPACECRAFT generate waste heat that is ultimately dissipated to deep space by infrared radiation. The simplest design to radiate

excess heat is to use part of the spacecraft's aluminum honeycomb wall, which is part of the spacecraft structure. Electronic components, which are the sources of the heat, are mounted on the inside of the panel [1]. The heat generated by the components is conducted through the honeycomb core and insert blocks to the outside radiator surface. The panel face sheet, typically made of aluminum or carbon-fiber-reinforced plastics, distributes the heat laterally over the radiator surface. Such passive structure radiators are conventional, and they are still common in uncrewed spacecraft.

When components cannot be mounted on the radiator panel, it is necessary to transport their waste heat to a thermal radiator. Heat pipes are most commonly used due to their lightness and high thermal conductance. Sometimes heat pipes are mounted inside the honeycomb core or on the panel face sheet to spread out the waste heat over the entire radiator surface for a uniform radiator surface temperature [2]. Since the 1990s, loop heat pipes [3] have been used to transfer heat to radiative areas of uncrewed spacecraft [4]. The evaporator of the loop heat pipe is mounted on the components, and the condenser is mounted on the radiator panel.

For crewed spacecraft, the heat rejection system has conventionally consisted of single-phase fluid loops and radiators. The heat generated by components are transferred to the radiator by mechanically pumped coolant through fluid loops and then emitted into deep space by the radiator. The pumped fluid loop system provides efficient transfer of a large amount of heat under a controlled heat transfer rate and operating temperature. It is suitable for a crewed mission that dissipates as much as several tens of kilowatts of heat, has a habitable module with a large volume where heat-generating components are distributed, and requires precise temperature control for humidity and air temperature. A shortcoming of the pumped fluid loop system is the potential failure of a mechanical pump and control electronics, but this can be repaired or replaced in orbit.

Multiple parallel flowpaths should be provided for a radiator panel of any heat rejection system using a mechanically pumped loop. Radiator panels are of two types, based on either a honeycomb structure or thin plates. The radiator panels of a space shuttle orbiter and the International Space Station (ISS) are representative of the honeycomb type. For both of these, the flow channels are embedded in an aluminum honeycomb core. The space shuttle orbiter has

radiators of two different thicknesses: the forward panels are 23 mm thick, and the aft panels are 13 mm thick. The panels have 0.28-mm-thick 2024-T81 aluminum face sheets and 5056-H39 aluminum honeycomb cores [5]. The silverized Teflon® tapes attached to the outer surfaces provide a thermal finish of high infrared emissivity and low solar absorptivity, and the parallel freon-21 coolant tubes are embedded in the core as shown in Fig. 1. To protect the tubes from micrometeoroids and debris (MMOD), 0.5-mm-thick aluminum doublers are added on the face sheet above the tube. Hyde et al. [5] reported that, over 122 shuttle flights, two perforations penetrated by MMOD were recorded; and none were on the tubes of the radiator panels.

The International Space Station has six deployed radiator arrays (radiator assemblies) that provide 70 kW heat rejection using two ammonia single-phase fluid loops. The radiators are designed for 10 years of use in the near-Earth-orbit space environment, including MMOD and solar radiation. A radiator assembly consists of eight identical radiator panels: each of which are 3.33 by 2.64 m. Twenty-two flow tubes are embedded in parallel for redundancy. Two supply and two return manifolds provide two independent flow passages [6]. The manifolds and flow tubes are made of Inconel 718 due to its high resistance to ammonia freezing and thawing pressures, as well as its corrosive resistance to ammonia. The flow tubes are bonded with silver-filled epoxy between two aluminum extrusions for MMOD protection. White paint is applied on the 0.025-mm-thick aluminum face sheets.

Radiator panels composed of thin aluminum plate instead of honeycomb are used by the heat rejection system (HRS) radiator of the Mars Pathfinder [7,8] and the Orion European Service Module (ESM) [9–11]. The Mars Pathfinder is the first uncrewed U.S. deep-space mission spacecraft that used an active pumped-liquid cooling system [1]. Its radiator is 8.22 m long by 0.2 m wide with a 0.75-mm-thick aluminum strip thermally attached and wrapped around 9.53-mm-diameter HRS tubing [7]. Both sides are painted white and reject up to 180 W of excess heat. The working liquid for the mechanically pumped loop is freon-11. Birur and Bhandari [8] reported that the HRS system functioned flawlessly during its entire seven-month flight from Earth to Mars.

NASA's Orion spacecraft will be supported by the ESA's ESM (a cylindrical 4.5-m-diameter unpressurized module) and will carry

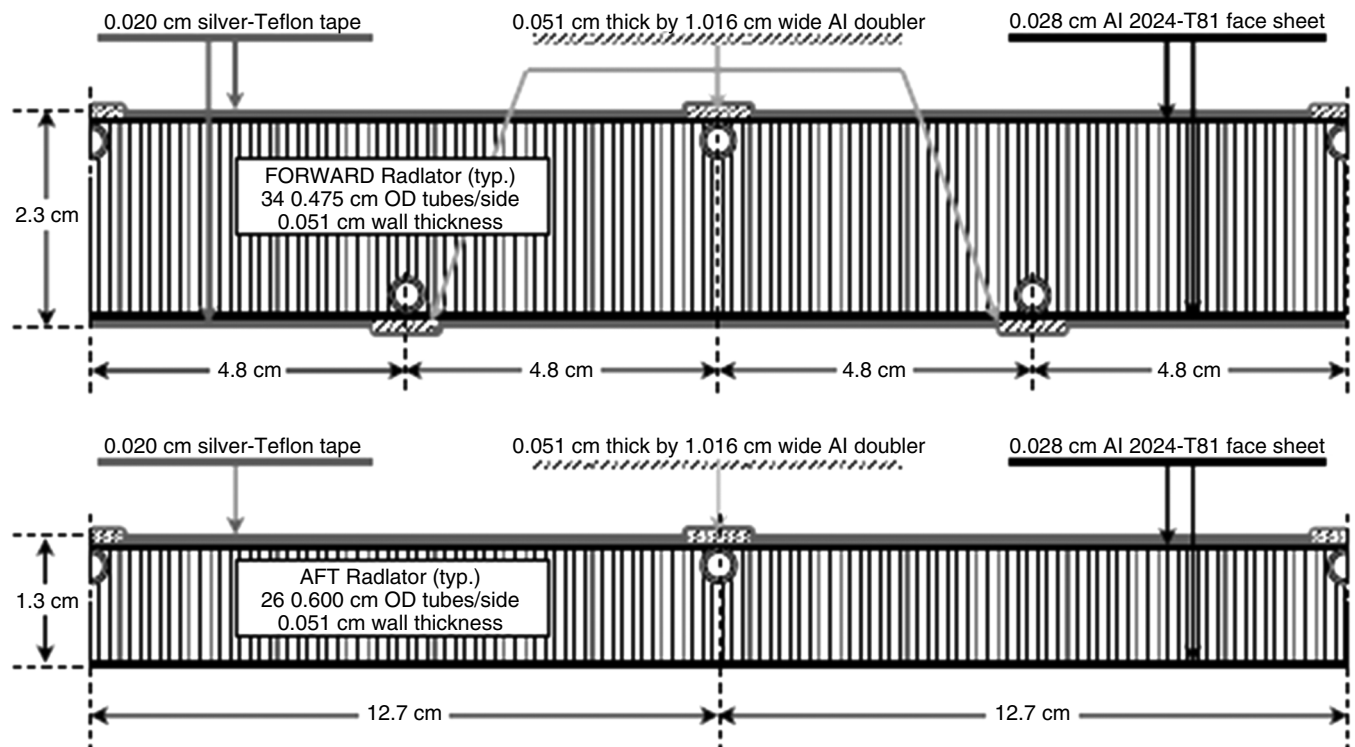


Fig. 1 Space shuttle radiator panel construction [5] (OD = outer diameter; incl. = inclination; typ. = typical configuration).

humans and cargo beyond low Earth orbit and return them safely to Earth [9]. A single-phase HFE7200 liquid loop transfers the heat from the ESM avionics and from the crew module to the ESM-mounted radiators [10]. The metal tubes' flowpaths are bonded to the radiators, which are curved plates forming an outer shell of the module and enclosing the ESM. The radiator panels are also used as bumpers to protect the ESM from MMOD impacts [11].

Recently, Hahn and Perich [12] presented an active thermal control system radiator for the Dream Chaser® cargo system. The radiator consists of an aluminum extrusion that contains a flow tube and a section of panel. The tube and panel are formed from the same piece of metal, and so there is no bond line. The extrusions are joined using friction stir welding. The design tradeoff between fluid tube sizing and the number of tubes was determined by thermal fluid analysis.

The fluid loop is still a common option for thermal control systems intended for future crewed missions such as the deep-space gateway [13] and for uncrewed spacecraft with a large heat dissipation of several kilowatts [14]. Japan developed single-phase water (inner loop) and FC-72 (exposed loop) mechanically pumped loops for the thermal control system of the ISS's Japanese experiment module, Kibo [15,16]. These loops have operated successfully for more than 10 years. Building on the experience of Kibo's mechanically pumped loop systems, the Japan Aerospace Exploration Agency (JAXA) has investigated a new active thermal control system (i.e., a mechanically pumped loop system) and developed a new mechanical pump for the future crewed missions [17].

The key requirements for radiator panels as part of a heat rejection system using fluid loops are high thermal performance (i.e., high fin efficiency), light weight, and adequate resistance against MMOD impacts. These are conflicting requirements, and a balance was found for the optimum design solution. This paper reports the new radiator design developed by JAXA. It is based on a typical aluminum honeycomb panel but designed with some ingenuity to reduce weight and improve thermal performance. A prototype radiator 0.97 m long and 0.79 m wide was built and tested with flowing single-phase HFE7200 liquid in a thermal vacuum chamber. Thermal Desktop and SINDA/FLUINT software were used to develop the thermal mathematical model, which was correlated with the test results and evaluated through thermal analysis.

II. Prototype Radiator Design Overview

A. Requirements

The requirements for the prototype radiator were defined as follows to be competitive with conventional space radiators using a single-phase pumped loop:

- 1) The maximum envelope had a length of 1100 mm, a height of 500 mm, and a width of 800 mm. (The envelope requirement is decided by the effective area of the thermal vacuum chamber.)
- 2) The fin efficiency η was greater than 0.9 and is defined in Eq. (1).
- 3) The area density (dry mass) was less than 7.0 kg/m² (target value).
- 4) For protection against MMOD impact, the flow tubes were required to last for 10 years in the assumed MMOD environment on an ISS orbit (400 km attitude, 51.6 deg inclination). The details will be given later in the paper.
- 5) The maximum design pressure (internal pressure) was 1 MPa (guaranteed pressure greater than 1.5 MPa, and burst pressure greater than 2.5 MPa).
- 6) The maximum leak rate was less than 1.7×10^{-5} standard cubic centimeters per second of helium gas (scsGHe) at the differential pressure of 7.1 kgf/cm².
- 7) For the thermal vacuum environment, the pressure less than 1.3×10^{-3} Pa and the temperature range was $-180 \sim 80^\circ\text{C}$.

The fin efficiency η is defined as follows:

$$\eta = \frac{Q_{\text{real}}}{Q_{\text{id}}} \quad (1)$$

Here, Q_{real} indicates the actual heat dissipation and Q_{id} is the identical heat dissipation calculated by assuming that the

radiator surface temperature is uniform at the maximum surface temperature.

Only the area density and fin efficiency are considered in comparing the radiator's performance to that of previous space radiators. The other requirements are related to its essential functions and the environment. Unfortunately, the area densities and fin efficiencies of past radiators cannot be found in the literature described in the previous section. A target fin efficiency of 0.9 was selected based on the authors' experience and typical values of radiators in Japanese uncrewed spacecraft (i.e., 0.8–0.9). The area densities of radiators for past and current crewed missions are taken from the literature [5,6,9–12]: 5.3 kg/m² for the space shuttle, 6.6 kg/m² for the ISS, 10.3 kg/m² for Orion, and 9.2 kg/m² for the Dream Chaser. These values were estimated from the dimensions and material specifications in the literature by dividing the panel area by the panel plus manifold masses (i.e., the total dry mass). These values are approximate because some information was not available from the literature, such as the adhesive mass (estimated by the authors based on a general panel type). Also, it should be noted that the area density for ISS radiators of 6.6 kg/m² is based on the area of both sides of the radiator, unlike the other radiators. According to Ref. [6], the flow channel is centered on both face sheets and there is an aluminum extrusion with a gap for MMOD protection between the channel and each sheet. Assuming that only one side of the panel faces deep space and one part of the extrusion is unnecessary, the area density is expected to decrease to around 6 kg/m².

The target area density of 7 kg/m² is less than those of recent aluminum-plate radiators but slightly more than those of the honeycomb type. The radiator mass is closely tied to the fin efficiency. Decreasing the flowpath pitch reduces the mass; however, it simultaneously causes a drop in the fin efficiency. The target value of 7 kg/m² was chosen as the minimum mass needed to keep the fin efficiency at 0.9; this mass can also be fabricated as a practical honeycomb panel.

B. Design Overview

The prototype radiator is composed of a honeycomb panel and provides good thermal and structural performance with low weight. The panel has 0.3-mm-thick A7075 aluminum face sheets and an A5052 aluminum honeycomb core. Figure 2 shows the dimensions and construction of the prototype radiator. Silverized Teflon® (0.127 mm thick) is attached to the outside of the panel, facing deep space, and has a highly infrared emissivity and low solar absorptive thermal finish. The other side of the panel is covered with multilayer insulation (MLI). There is one supply manifold along one edge of the panel; the return manifold is located along the other side. The inside diameter of the manifolds is 12 mm. Eight flow tubes with an inner diameter of 5 mm are joined to the two manifolds at a 90 mm pitch. Both the manifolds and tubes are made of A5052 aluminum. All joints of tubes and manifolds are soldered as shown in Fig. 3. The outer and inner surfaces of tubes and manifolds are coated with

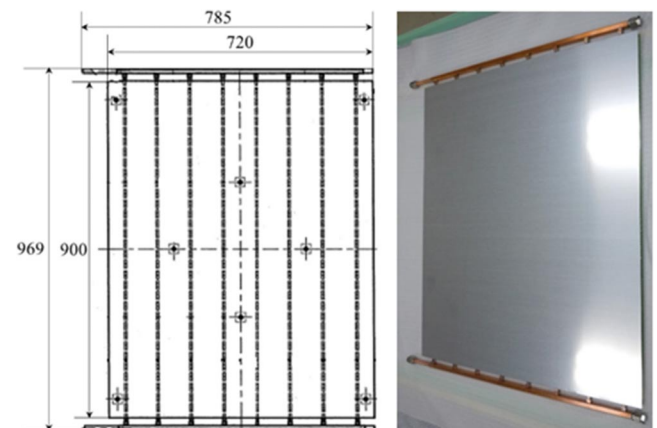


Fig. 2 Prototype radiator (units: millimeters).

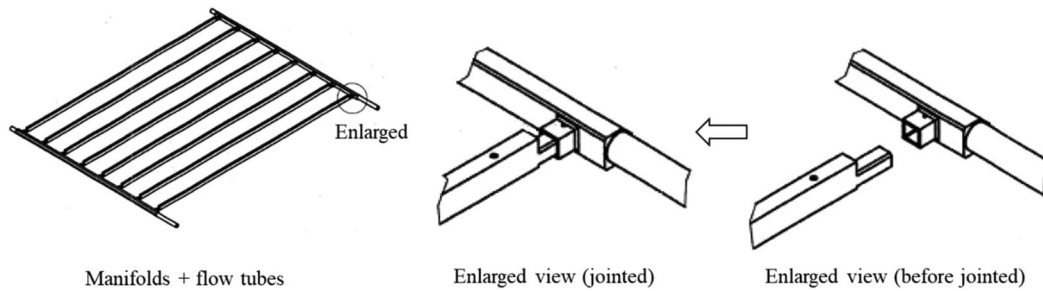


Fig. 3 Joint of tube and manifold.

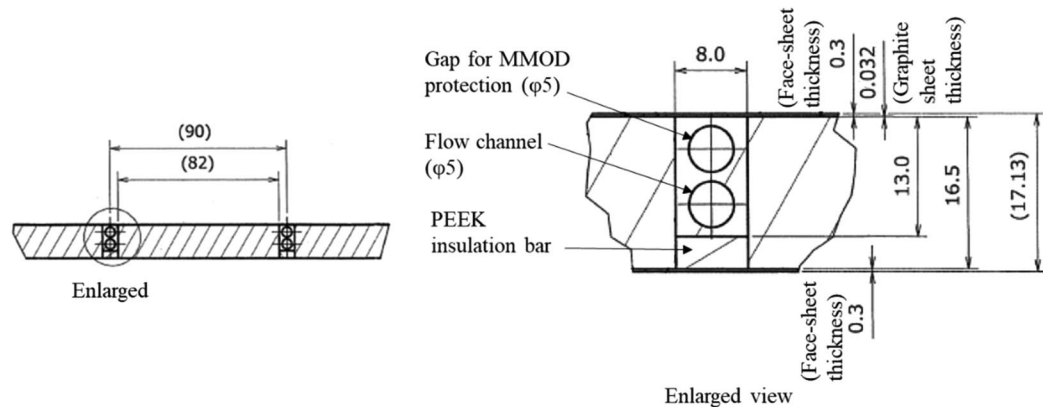


Fig. 4 Cross section of prototype radiator.

copper to improve wettability for the solder. To prevent the radiator from freezing if the heat load is lost, a heating tape is attached to both manifolds.

The eight flow tubes are located between the two face sheets. A cross section of the flow tubes is shown in Fig. 4. Each honeycomb cell is perforated and has a 6.25 mm (1/4 in.) width, which are 0.0254 mm (0.001 in.) thick. There is a 5 mm gap above the flow tube for MMOD protection. The gap and flow tube (channel) are machined from a single aluminum bar that is 8 mm wide and 13 mm thick. This provides good thermal conductivity between the inner wall of the flow tube and the outer surface of the panel. A thermal insulation bar made of Poly Ether Ether Ketone (PEEK) is inserted between the flow tubes and the face sheet on the bottom side to prevent heat transfer from the coolant to the surface that is covered with MLI. The insulation bar is the same length as the widest part of the panel. To reduce the mass and undesired heat flow, the bar has 89 cylindrical holes that are 5 mm in diameter. The insulation bars and aluminum bars where flow channels and gaps are machined are bonded to the core with a foam adhesive.

GRAPHINITY™, which is an ultrahigh thermally conductive graphite sheet (thickness: 0.032 mm) is attached to the back (i.e., the honeycomb core side) of the aluminum face sheet to improve in-plane heat conduction and fin efficiency. The density and in-plane thermal conductivity of the graphite sheet are 2000 kg/m³ and 1500 W/m/K, respectively. The two face sheets are bonded to the honeycomb core with a sheet adhesive. To prevent separation of the face sheet and the core due to delamination of the graphite sheet, the graphite sheet is smaller than the face sheet and the sheet adhesive by 20 mm at each end. Sample sheets did not separate in thermal cycle testing, which ran between room temperature and liquid nitrogen temperature.

C. MMOD Protection

Meteoroid velocities range from 11 to 72 km/s with an average of 22–23 km/s [18]. A meteoroid density distribution based on a meteoroid radar survey shows the bulk density ranging from 0.1 to 2.5 g/cm³ or more, and the mean bulk densities determined by

three-dimensional spherical and overdense theories were 0.6 ± 0.5 and 0.7 ± 0.6 g/cm³ [19]. The impact velocities of orbital debris are lower than those of meteoroids, generally striking spacecraft in low Earth orbit at relative speeds of less than 1 km/s to over 15 km/s, with an average velocity of about 9 km/s at a 400 km altitude. For debris risk assessments, the particle density of orbital debris is typically assumed to be 2.8 g/cm³ [20]. Regarding the particle size, the cross-sectional flux is predicted by the NASA orbital engineering model, ORDEM2000, as shown in Fig. 5.

Based on the preceding information, the radiator was designed so that the flow tubes will not be damaged, even by an impact with up to the following MMOD description: 1) meteoroid with a density of 0.7 g/cm³, a velocity of 23 km/s, a diameter of 0.6 mm, and an impact angle of 0 deg (normal impact) and 2) orbital debris with a density of 2.8 g/cm³, a velocity of 9 km/s, a diameter of 0.9 mm, and an impact angle of 0 deg (normal impact).

The following equations can be used in the preliminary design of MMOD protection using Whipple shields. A Whipple shield,

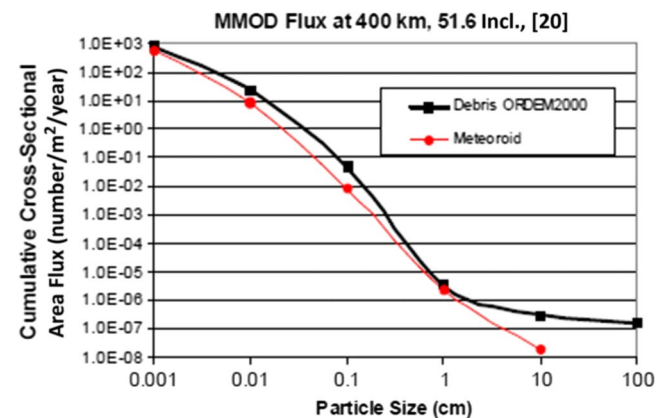


Fig. 5 Flux of meteoroids and orbital debris in low Earth orbit in 2006 [20].

invented by Whipple [21], consists of a relatively thin outer bumper that is separated from the main structure (rear wall) and disperses the kinetic energy of the incoming MMOD. Assuming that the normal component of the projectile velocity is 7 km/s or more, the bumper and rear wall thickness to defeat a given threat particle are determined by Eqs. (2) and (3), respectively [20]:

$$t_b = \frac{c_b m_p}{\rho_b} = \frac{c_b d_p \rho_p}{\rho_b} \quad (2)$$

$$t_w = c_w d_p^{0.5} (\rho_p \rho_b)^{1/6} (M_p)^{1/3} V_n / S^{0.5} \left(\frac{70}{\sigma_y} \right)^{0.5} \quad (3)$$

where c_b is a coefficient assumed to be 0.25 when S/d is less than 30. Another coefficient, c_w , is given as $0.16 \text{ cm}^2 \cdot \text{s} / (\text{g}^{2/3} \cdot \text{km})$ [20].

From the preceding equations, the minimum thickness of the bumper and rear wall that meet the requirements for MMOD protection are calculated to be 0.2 and 0.9 mm, respectively. Due to manufacturing constraints, the prototype radiator has a minimum bumper thickness of 1.3 mm and a minimum rear wall thickness of 1 mm.

D. Compliance with Requirement

The prototype radiator was inspected to check compliance with the requirements of envelope, area density, pressure resistance, and minimum leak rate. It was confirmed that all requirements except area density were satisfied. The area density is 7.1 kg/m^2 , which deviates from the target value by 0.1 kg/m^2 but is comparable to other radiators being developed. As mentioned in Sec II.A, the mass correlates with fin efficiency. Based on design calculations, reducing the number of flow channels by one decreases the area density to 6.7 kg/m^2 and reduces the fin efficiency to less than 0.9. Section IV.C discusses suggestions for improving the area density in ways other than the number of flowpaths. The fin efficiency and the thermal vacuum resistance were verified by a thermal vacuum test.

III. Thermal Vacuum Test

To verify the heat rejection performance capability and thermal vacuum resistance of the prototype radiator, a thermal vacuum test was conducted at the JAXA Tsukuba Space Center.

A. Test Setup

The system test layout and other single-phase fluid loop test hardware are shown in Fig. 6. Only the prototype radiator, plumbing, and temperature sensors were installed in the vacuum chamber as shown in Fig. 7; the other hardware was located outside the chamber. The chamber's liquid nitrogen shroud and the door were cooled to the temperature of liquid nitrogen (approximately 100 K) to simulate a deep-space environment. The radiator panel was installed horizontally in the chamber secured by eight 3-mm-thick aluminum plate jigs. A glass epoxy plate is sandwiched between the aluminum plate jig and the radiator panel to insulate them. The top side of the panel is coated with silverized Teflon® and radiatively coupled to the black-painted shroud surface. The bottom of the panel is covered with MLI to insulate the shroud. The plumbing and two manifolds are also covered with MLI.

The testing was done on a circulation system in which the coolant (3M HFE7200), heated by two heater plates, flowed into the radiator. Then, the coolant was cooled and returned to the heater plate. The coolant was circulated by a stainless-steel magnet-drive centrifugal pump (Sanwa MML pump) that can be used down to a coolant temperature of -80°C . The flow rate of HFE7200 through the radiator and the heater plates is controlled manually by adjusting the opening of manual valves V3 and V4 (see Fig. 6), which are located downstream of the pump. The heater plate is made of copper plate that is 12 mm thick, 120 mm long, and 120 mm wide. A flow

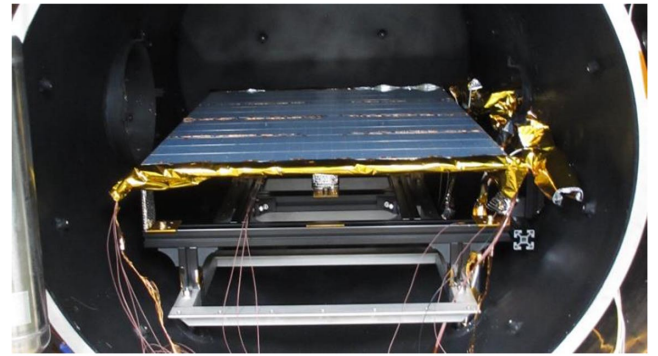


Fig. 7 Prototype radiator installed in vacuum chamber.

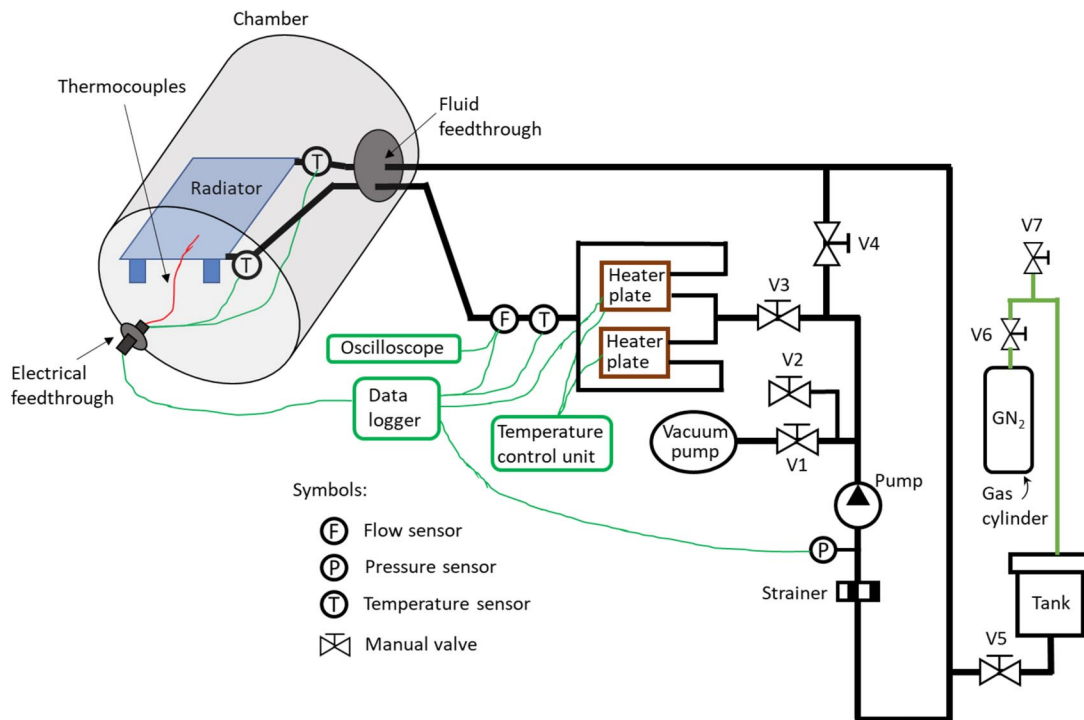


Fig. 6 Test layout.

tube, two cartridge heaters, and a K-type thermocouple are inserted so that the temperature of the plate can be kept constant while the coolant circulates. Since HFE7200 is a highly volatile coolant, the test was conducted under a nitrogen pressure of 1.2 atm with HFE7200 stored in a tank, which is upstream of the pump and is pressured by a nitrogen gas cylinder (capacity: 47 liters, charging pressure: 14.7 MPa). In other words, the tank works as an accumulator, whereas it does not have a bellows for a gas–liquid separation. All plumbing, valves, and heater plates exposed to the coolant are isolated from their surroundings by heat insulation.

Fifty-two T-type thermocouples are attached to the silverized Teflon® surface of the radiator panel, and nine thermocouples are attached to the other surface, as shown in Fig. 8, in order to measure its temperature gradient. One T-type thermocouple is attached to each outer surface of the inlet and outlet manifolds. An ultrafine thermocouple with a core wire that is 0.1 mm in diameter and has a polyimide coating that is 0.005 mm thick is used for the radiator to minimize heat leaks and the noneffective field of view due to the thermocouples. To measure the coolant temperature, three temperature sensors (Amphenol GE-1920) are positioned at the inlet and the outlet of the manifolds and downstream of the heater plates, as shown by the capital letter T in Fig 6. A wheel flowmeter (Tokyo Keiso W-2013N) is mounted downstream of the heater plates and upstream of the radiators to measure the flow rate of the coolant through the radiator. A pressure sensor (Kulite X-140M-35BARA) is in the pipe downstream of the pump to measure the discharge pressure. All the data measured by the aforementioned thermocouples and the sensors are acquired by a data logger (Graphtec GL7000), and the flow meter data are monitored on an oscilloscope (Tektronix MDO 3024). Throughout the test, all data of the tested hardware were recorded continuously on the data logger with a sampling rate of 1 Hz.

Before the chamber door was closed, the radiator to be tested was filled with HFE7200 by the following procedure:

- 1) The tank is filled with HFE7200, and V5 is closed.
- 2) The plumbing, manifolds, and flow tubes of the radiator are evacuated; V1, V3, and V4 are opened.
- 3) V1 and V2 are closed.
- 4) V7 is closed and V6 is opened until secondary nitrogen gas (GN2) pressure reaches about 1.2 atm.
- 5) V5 is opened, and the tested equipment is filled with HFE7200.
- 6) V2 is opened, and HFE7200 is checked to ensure that it is flowing out sufficiently to confirm that the tested hardware is completely filled with HFE7200.

Here, V1–V7 correspond to the manual valves shown in Fig. 6. To take advantage of gravity, the tank is located at the highest position in the tested equipment and V2 is in the highest position. After

confirming that there are no leaks of 1 ppm or more into the atmosphere and that the chamber is using a portable refrigerant monitor (Bacharach PGM-IR), the pump was rotated to perform a trial run. The chamber door was closed, and the evacuation was started after the trial run was completed.

B. Test Conditions

The tests were conducted by varying the flow rate and the temperature of the heater plate as shown in Table 1. In all cases except for case 13, the data were acquired under the same test conditions until a thermal equilibrium was reached. The criterion of the thermal equilibrium is that the temperature change rate at each measurement point of the radiator panel is 0.3°C/h or less. The pump was stopped for case 13 to measure the rate of decrease in the radiator temperature. In all test cases, the shroud temperature and the chamber pressure were kept at around 100 K and less than 10^{−3} Pa, respectively.

C. Modeling

A thermal mathematical model was created to perform thermal analysis for simulating the thermal vacuum test. A finite difference and lumped parameter tool (SINDA/FLUINT version 6.1) was used as a solver to analyze thermodynamic and hydrodynamic behavior by network analogy. Thermal Desktop version 6.1, which is a CAD-based modeling builder, was used to create the nodes and the conduction and radiation network. Thermal Desktop’s software module

Table 1 Test conditions

Case no.	Flow rate, kg/h ^a	Heater plate control temperature, °C
1	101	18
2	101	8
3	101	−2
4	53	−2
5	53	8
6	53	18
7	201	18
8	201	15
9	201	8
10	201	6
11	201	−2
12	201	−3
13	201–0 (for 10 min)	−10 (heater off)

^aThe mass flow rate of the coolant flowing into the radiator is shown in the flow rate column.

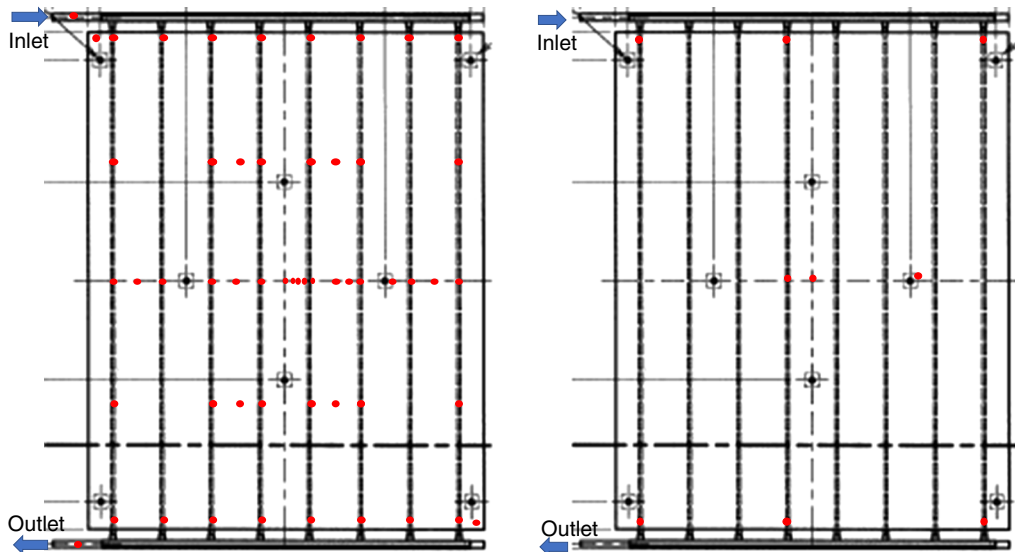


Fig. 8 Temperature measurement points: silverized Teflon® side (left), and MLI side (right).

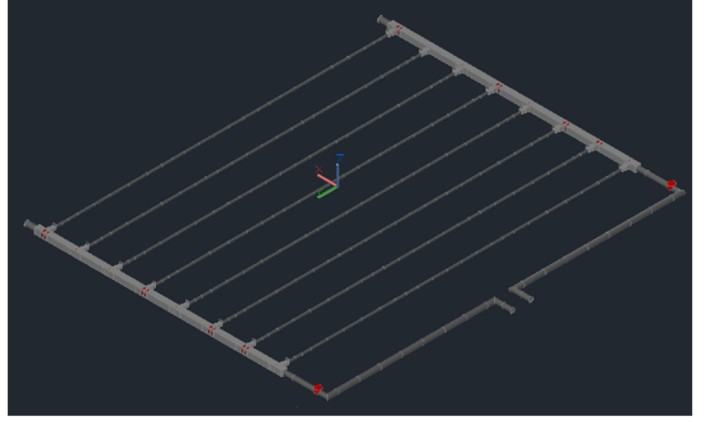
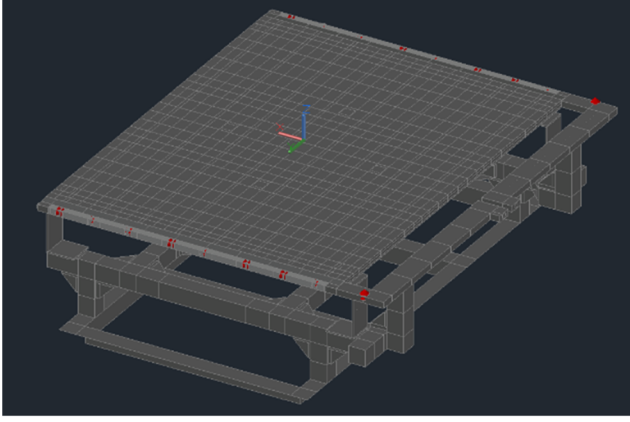


Fig. 9 Thermal mathematical model of prototype radiator mounted on jigs.

FLOCAD was also used to build a geometry-based hydrodynamic model. The model consists of 6540 thermal nodes and 302 fluid lump nodes. The models are shown in Fig. 9.

A whole model, except a vacuum chamber, is shown in the left-side figure, and a flow model extracted from the whole model is shown in the right-side one.

In this model, a convection heat transfer rate between the coolant and the tube wall h is derived as follows:

$$h = \frac{Nu \times k}{d_i} \quad (4)$$

Here, k is the thermal conductivity of the coolant and d_i is the tube diameter. The Nusselt number Nu is calculated, depending on the flow condition (i.e., the Reynolds number Re). For fully developed laminar flows ($Re \leq 2300$), the Nusselt number Nu_{lam} is calculated from an approximate equation based on the literature [22]. Here, the flowpath is assumed to be a concentric annular duct, as shown in Fig. 10. The actual flowpath is not exactly in the shape of a concentric annular duct, but it is the closest approximation to predict the Nusselt number. The wall heat fluxes on the inner and outer walls are assumed to be constant at any axial location but different. The peripheral wall temperature gradient is constant. Shah and London [22] proposed the Nusselt number for the inner wall in this case Nu_{lam} as a value that depends on the ratio of the outer radius to the inner radius r^* , as Fig. 10 shows. The power approximation for the plots in Fig. 10 is as follows:

$$Nu_{lam} = 7.985r^{*-0.356} \quad (5)$$

Here, r^* is the ratio of the outer radius and inner radius of the duct:

$$r^* = \frac{r_i}{r_o} \quad (6)$$

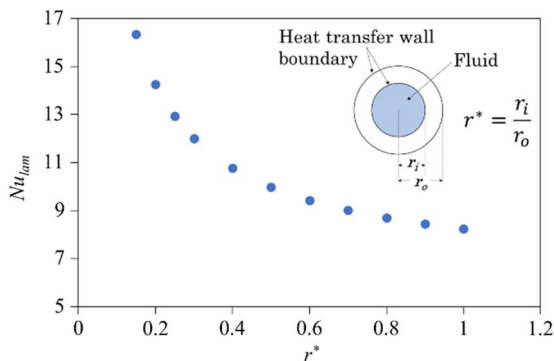


Fig. 10 Nusselt number cases for concentric annular duct for fully developed laminar flow [22].

For transitional flows ($2300 < Re \leq 3000$), the Nusselt number Nu_{tra} is calculated by Hausen's correlation [23,24].

$$Nu_{tra} = 0.116(Re^{\frac{2}{3}} - 125.0)Pr^{0.3} \quad (7)$$

For turbulent flows ($Re > 3000$), the Nusselt number Nu_{tu} is calculated by the Dittus–Boelter correlation [24,25]:

$$Nu_{tu} = 0.023Re^{0.8}Pr^{0.3} \quad (8)$$

The model is validated by the test results. Figure 11 shows a histogram of the radiator panel temperature difference for 12 thermal equilibrium cases, totaling 930 temperature measurement points. More than 98.8% of the predicted temperatures are within $\pm 5^\circ\text{C}$ of the measured temperature and 46.2% are within $\pm 1^\circ\text{C}$ of that. The analysis was also performed at high and middle mass flow rates but was not included under the thermal vacuum test conditions (150, 250, and 300 kg/h).

IV. Results and Discussion

A. Heat Rejection Performance

The radiator's performance in rejecting the heat in a simulated deep-space environment was measured in the tests and analyses. Figure 12 shows the measured and predicted temperature gradient of the radiator surface for test cases 1, 2, and 3. In these cases, the coolant mass flow rate was constant at 101 kg/h; and the inlet temperature was changed to 18°C (case 1), 8°C (case 2), and -2°C (case 3). The maximum uncertainty of measured temperature including the standard deviation of the arithmetic mean is 1.6%. The temperature gradient of the radiator surface is within 10°C , and the gradient does not change under the same mass flow rate condition if the inlet temperature changes. The four points where the temperature is 2 to 3°C lower than the surrounding area are at the jig mountings.

Predicted temperatures are shown in contour plots, and the color legend is shown to the left of each plot. The measured temperatures are shown as black dots, and values in the figure are the averages over 1 h under thermal equilibrium for each case. (The same applies to Figs. 13 and 14.)

Figure 13 shows the measured and predicted temperature gradients of the radiator surface for test cases 2, 5, and 9. In these cases, the coolant inlet temperature was constant at $7 \pm 1^\circ\text{C}$; and the mass flow rate varies from 53 kg/h (case 2), 101 kg/h (case 5), and 201 kg/h (case 10). The coolant temperature and mass flow rate for the same cases are shown in Figs. 14 and 15, respectively. The analysis and test results show that the radiator panel temperature at the most downstream part is 1 to 2°C higher than at 20 cm upstream. On the other hand, the predicted coolant temperature becomes lower toward the downstream side; and there is no tendency like the panel temperature. It is assumed that the radiator panel at the most downstream side is heated due to heat leaking from the manifold. As

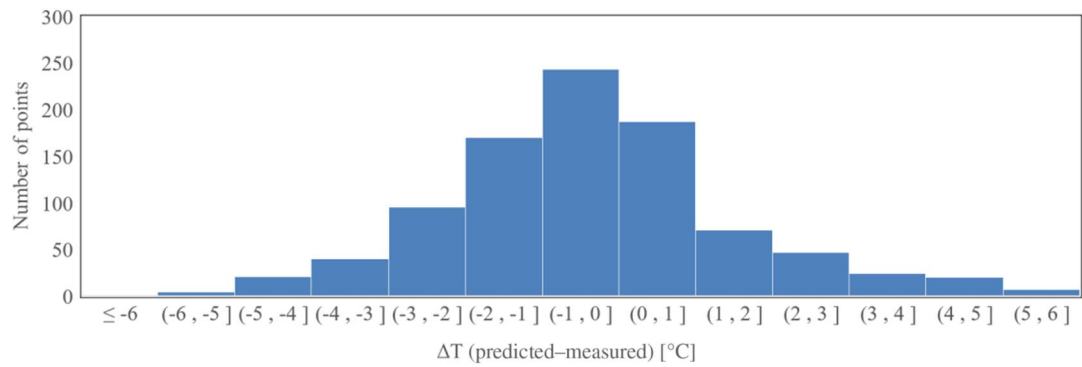


Fig. 11 Histogram of difference between predicted and measured temperatures.

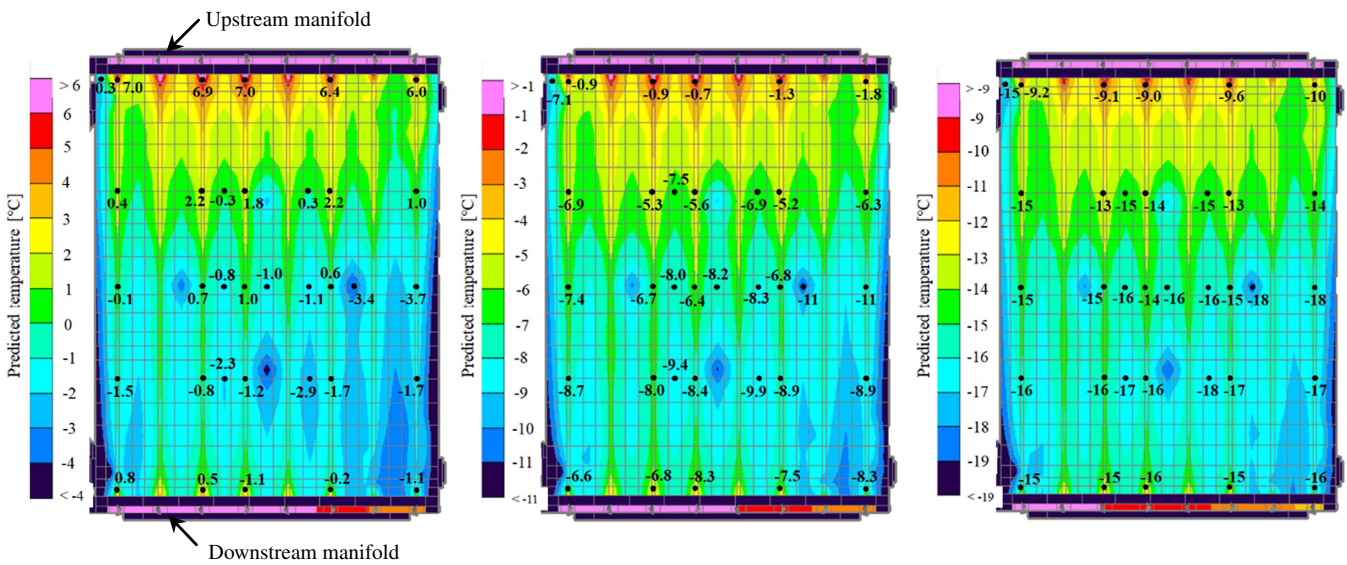


Fig. 12 Measured and predicted radiator temperatures of cases 1, 2, and 3: case 1 (leftmost), case 2 (center), and case 3 (rightmost).

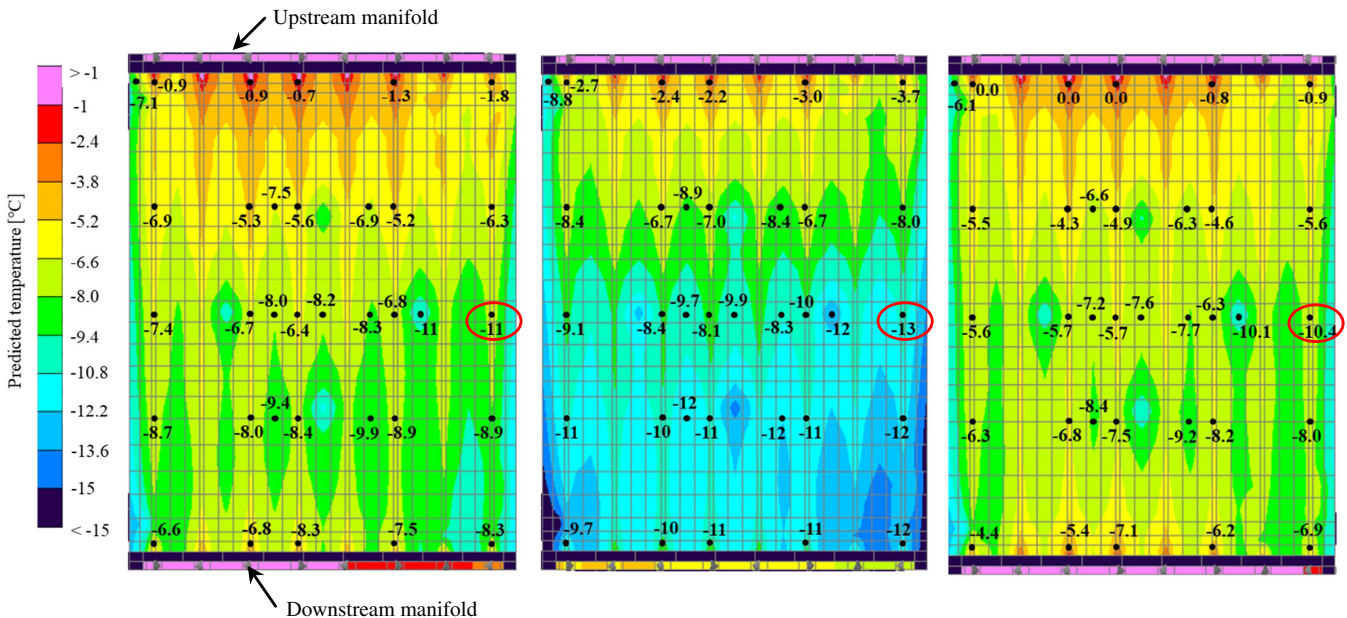


Fig. 13 Measured and predicted radiator temperatures of cases 2, 5, and 10: case 2 (leftmost), case 5 (center), and case 10 (rightmost). (The same applies to Fig. 14 and 15.)

Figs. 12 and 13 show, the temperature of the manifolds is higher than that of the radiator panel, because the manifolds are covered with MLI and are thermally insulated from LN2 shroud.

The test and analytical results are in good agreement: within temperature differences of 5°C . However, the predicted temperature near the center just above the right-side flow channel (indicated by a

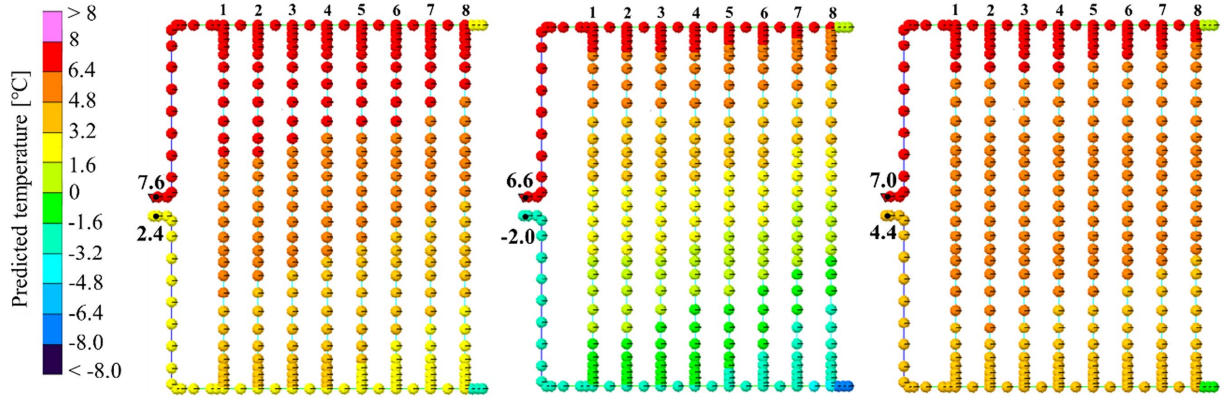


Fig. 14 Measured and predicted coolant temperatures of cases 2, 5, and 10.

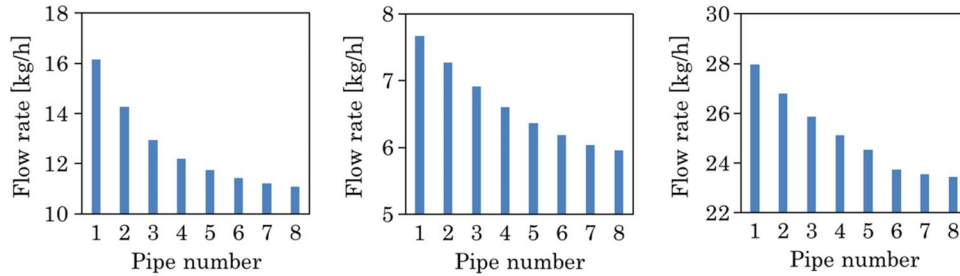


Fig. 15 Predicted mass flow rates of cases 2, 5, and 10. Pipe numbers are indicated in order of proximity to the upstream and high-temperature side as shown in Fig. 14.

red circle in Fig. 13) does not match the measured temperature. Unlike the trends of the other measured temperatures above the flow channel and the predicted temperature, the indicated measured temperature is lower than the downstream temperature. This is because the flow rate in the rightmost flow channel is less than in the others. The analysis results also show a variation, and the flow rate at the flow channel farthest from the manifold inlet has a lower flow rate than other parts, as shown in Fig. 15. However, it is assumed that the nonuniformity of the actual flow is stronger than that in the model and that the actual flow rate of the farthest channel is lower than that in the model. The low flow rate emphasizes the heat leaking to the mounting jig, and it causes a lower temperature than downstream.

At a mass flow rate of 50 kg/h, the coolant temperature from the inlet to the outlet dropped more than 8°C, whereas it is lower than 3°C at a mass flow rate of 201 kg/h. It is caused by the lower flow velocity. The temperature gradient in the radiator panel surface is also larger as the mass flow rate is lower.

Fin efficiency is calculated from the test and analysis results by Eq. (1). The actual heat dissipation rate for the analysis is calculated from the total amount of heat dissipation derived from the area and temperature of each node:

$$Q_{\text{real}_a} = \sum_i A_i \sigma_s \epsilon (T_i^4 - T_{\text{sh}}^4) \quad (9)$$

Here, A_i and T_i are the area and temperature of each node, σ_s is the Stefan–Boltzmann constant, ϵ is the hemispherical infrared emissivity (0.78), and T_{sh} is the shroud temperature. The actual heat dissipation rate for the test cases was calculated by a method different from the analysis because the surface temperature gradient could not be measured as finely as in the analysis. It is determined by subtracting the amount of heat leaked from the radiator to the surroundings from the amount of heat exchanged between the cooling water passing through the radiator and the surroundings. It is expressed as the following equation:

$$Q_{\text{real}_a} = \dot{m} C_p (T_{\text{in}} - T_{\text{out}}) - Q_{\text{leak_MLI}} - Q_{\text{leak_mnf}} - Q_{\text{leak_jig}} - Q_{\text{leak_side}} \quad (10)$$

Here, $Q_{\text{leak_MLI}}$ is the heat leakage through the MLI attached to the back of the radiator panel, $Q_{\text{leak_mnf}}$ is the heat leakage through manifold MLI, $Q_{\text{leak_jig}}$ is the heat leakage through the four mounting jigs, and $Q_{\text{leak_side}}$ is the heat leakage through the four sides of the radiator panel. The ideal heat dissipation is calculated by the radiator area A_{rad} , ϵ , and the maximum surface temperature T_{max} :

$$Q_{\text{id}} = A_{\text{rad}} \sigma_s \epsilon (T_{\text{max}}^4 - T_{\text{sh}}^4) \quad (11)$$

For a test case, T_{max} is the maximum surface temperature of the measured average temperature for 1 h at thermal equilibrium for each case. For the analysis, it is treated as the maximum predicted surface temperature. The fin efficiencies for the analysis and test cases are shown in Fig. 16. The fin efficiency of the test cases meets the requirement (greater than 0.9), except for the case with the lowest mass flow rate. In the low flow rate region, the radiator temperature decreases more quickly as the coolant flows downstream; and then it causes a smaller heat rejection and lower fin efficiency than that in a higher flow region. It

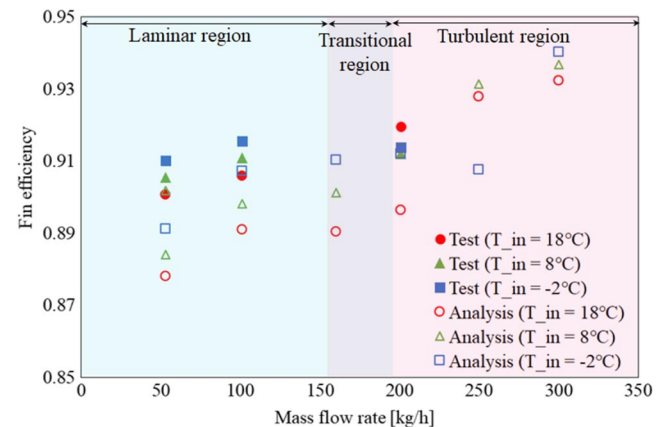


Fig. 16 Fin efficiency. The inlet coolant temperature is shown as “T_in” in the legend.

is necessary to increase the mass flow rate to the turbulent flow region to obtain a fin efficiency of 0.92 or more. Whether a high mass flow rate can be obtained depends on the operating range of the pump and the electric power allocated to the pump.

B. Transitional Temperature Change When Pump Stopped

The measured and predicted transitional temperatures when the pump stopped are shown in Fig. 17. The analysis was performed until steady state was reached, whereas the test was conducted for only 10 min. For the first 10 min, the predicted temperature agreed well with the difference from the measured temperature: within 1.5°C. In the thermal analysis, the coolant temperature is calculated to be constant at the freezing point of HFE7200 (−138°C) after it reaches that point. The coolant temperature reached that point 470 min after the pump stopped. Therefore, if the pump stops in deep space, it assumes that the coolant inside the radiator has frozen unless the pump is restarted within 7.8 h or the radiator receives a sufficient heat load from the sun, the Earth, or the electrical heaters.

The temperature at the center of the radiator surface is indicated as the panel temperature. The coolant temperature shows the predicted temperature of the coolant flowing in the fourth flow channel from the left at an intermediate point between the upstream and downstream locations.

C. Future Design Optimization

The analysis shows that the turbulent flow leads to high fin efficiency. A simple measure to reach is increasing the mass flow rate, but it can be constrained by mechanical pump performance or the electrical power budget. Another possibility is to increase the friction between the coolant and the wall to transit the flow to a turbulent flow, enhancing the wall convection heat transfer coefficient. Numerous studies of single-phase flow heat exchangers have proposed several additional processing measures, such as inclined grooves [26], twisted tape insertion [27], and a roughened micro-pin-fin wall [28]. It is expected that the fin efficiency will be improved even at a low mass flow rate by performing such processing on the flow channels, although this is a tradeoff among the performance, mass, and cost.

The following designs are proposed to further reduce the mass of the prototype without compromising the thermal performance:

1) Change the material of the insulation between the flow channel and aluminum skin from PEEK to the aluminum honeycomb core. This could not be realized with the prototype due to manufacturing difficulties and time constraint, but the area density can be reduced to 0.3 kg/m², as design a in Table 2 shows.

2) Reduce the channel diameter. If the channel diameter is decreased from 5 to 3 mm and the width of the aluminum bar is

reduced from 8 to 6 mm, the area density is further reduced by 0.4 kg/m² from that of design a. This approach also enhances the fin efficiency because the flow velocity increases under the same total mass flow rate. For example, with a total mass flow rate of 100 kg/s, the 3-mm-diameter channel develops turbulent flow, whereas the 5-mm-diameter channel has a transitional flow.

3) Reduce the adhesive mass. The adhesive mass used for the prototype is estimated at 1.1 kg, which is one-quarter of its total mass. If the adhesive mass can be reduced to 70% of the prototype, the area density can be further reduced by 0.5 kg/m² from that of design b, and eventually 5.7 kg/m², which is close to the estimated 5.3 kg/m² of a space shuttle orbiter.

V. Conclusions

A honeycomb panel radiator was designed for a single-phase pumped fluid loop, and a prototype has been built. The prototype radiator assembly consists of a honeycomb radiator of approximately 0.65 m², eight aluminum pipes inserted through the honeycomb core, and two aluminum manifolds joined at each end of the eight pipes. A gap serving as protection against micrometeoroids and orbital debris is provided between the radiator surface and the flow channels. The area density of the radiator is measured to be 7.1 kg/m² with the two manifolds and without the coolant. It was confirmed that the prototype radiator meets the requirements of pressure resistance and minimum leakage rate.

A thermal vacuum test was conducted to evaluate the thermal performance of the radiator. The radiator was installed in the vacuum chamber where the shroud was cooled with liquid nitrogen. The flow rate of HFE7200 coolant was changed from 50 to 200 kg/h. More than 50 temperatures on the radiator surface were measured at each case. Thermal analysis was also performed with Thermal Desktop and SINDA/FLUINT software. The thermal mathematical model was validated by the test results. The fin efficiency of the radiator increases as the coolant mass flow rate increases. A fin efficiency of 0.9 or higher was obtained from the test results.

Acknowledgments

This work was funded by Japan Aerospace Exploration Agency. The authors wish to thank Yukihiro Ueda of IHI AEROSPACE Company, Ltd. for the great advice on the setup and condition of the thermal vacuum test.

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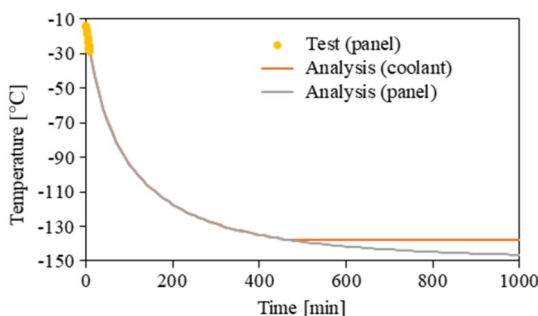


Fig. 17 Transitional temperature of case 13.

Table 2 Mass reduction potential

Design	Area density, kg/m ²
Prototype	7.1
a (change insulation material)	6.8
b (design a + reduce flow channel diameter)	6.2
c (design b + reduce adhesive mass)	5.7

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